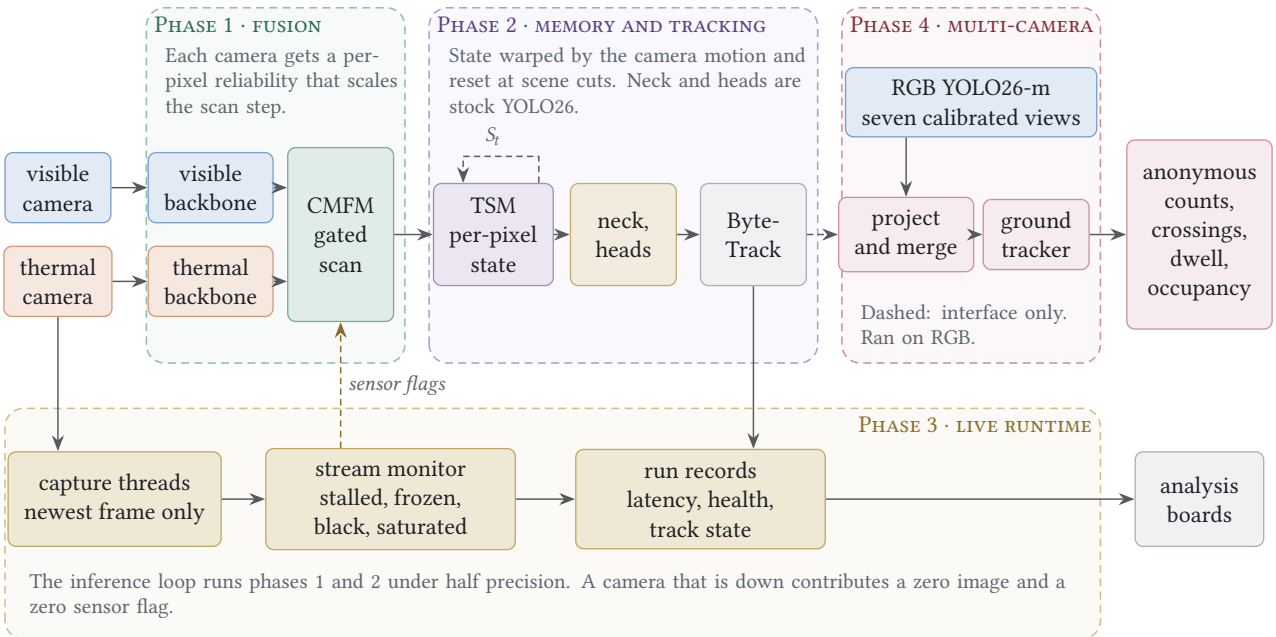
Outputs
Phase 1 · fusion
Each camera gets a per-pixel reliability that scales the scan step.
Phase 2 · memory and tracking
State warped by the camera motion and reset at scene cuts. Neck and heads are stock YOLO26.
S_t
Phase 4 · multi-camera
RGB YOLO26-m
seven calibrated views
visible camera
thermal camera
visible backbone
thermal backbone
CMFM gated scan
TSM per-pixel state
neck, heads
Byte-Track
project and merge
ground tracker
anonymous counts, crossings, dwell, occupancy
Dashed: interface only. Ran on RGB.
sensor flags
Phase 3 · live runtime
capture threads newest frame only
stream monitor stalled, frozen, black, saturated
run records latency, health, track state
analysis boards
The inference loop runs phases 1 and 2 under half precision. A camera that is down contributes a zero image and a zero sensor flag.